SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA1610 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.
Assessment Tool Weightage: 40%

QUESTIONS: 05 **Total Marks:50**

Annexure A

Industry Problem Solving Task

Criteria	Understanding of Industry Problem *(2)*	Application of Engineering Concepts *(2.5)*	Solution Design *(2.5)*	Use of Tools *(2)*	Analysis *(1)*	***Total(10)***
Weightage	*20%*	25%	25%	20%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I _P. Madesh_192365007_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

P. Madesh

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Annexure A

Q1 . Dataset: Supermarket Purchases

Customer	Visits/Month	Avg Spend (₹)	Items Bought
C1	10	2000	15
C2	4	5000	8
C3	12	1800	18
C4	3	6000	6
C5	11	2100	16

Question:
Using the above dataset, propose a clustering approach to segment customers and explain how it supports targeted marketing. *(BL3 – 7 Marks)*

Q2. Dataset: Credit Card Transactions

Transaction	Amount (₹)	Location	Time (hrs)
T1	200	Chennai	10
T2	5000	Delhi	2
T3	150	Chennai	11
T4	7000	Mumbai	1
T5	180	Chennai	12

Question:
Design a clustering-based solution using this dataset to detect unusual or fraudulent transactions. *(BL4 – 7 Marks)*

Q3. Dataset: Telecom Usage

User	Call Duration (min)	Data Usage (GB)	Recharge (₹)
U1	300	5	200
U2	100	2	100
U3	350	6	220
U4	80	1	90
U5	320	5.5	210

Question:
Explain how clustering can be applied to identify churn-prone users and improve customer retention strategies. *(BL4 – 7 Marks)*

Q4 . Dataset: E-commerce Behavior

User	Product Views	Clicks	Purchases
U1	50	20	5
U2	30	10	2
U3	60	25	6
U4	20	8	1
U5	55	22	5

Question:
Suggest a clustering technique to group users and explain how it improves recommendation systems. *(BL3 – 7 Marks)*

Q5. Dataset: Traffic Data

Location	Vehicle Count	Avg Speed (km/h)	Delay (min)
L1	200	30	10
L2	500	15	25
L3	220	28	12
L4	600	10	30
L5	210	32	9

Question:
Design a clustering-based solution to identify traffic congestion zones and explain how it helps in traffic management. *(BL4 – 7 Marks)*

Q1. Customer Segmentation for Targeted Marketing — Supermarket Purchases

Question: Using the above dataset, propose a clustering approach to segment customers and explain how it supports targeted marketing. (BL3 – 7 Marks)

1. Dataset

Customer	Visits/Month	Avg Spend (₹)	Items Bought
C1	10	2000	15
C2	4	5000	8
C3	12	1800	18
C4	3	6000	6
C5	11	2100	16

2. Understanding the Industry Problem

A supermarket wants to move from mass, one-size-fits-all promotions to targeted marketing. The business needs to know which customers behave alike in terms of how often they shop, how much they spend per visit, and how many items they buy, so that offers (loyalty points, bulk discounts, premium bundles) can be matched to each group instead of blasted to everyone. This is a classic RFM-style (Frequency, Monetary, Basket-size) customer-segmentation problem.

3. Proposed Clustering Approach

K-Means clustering is proposed as the primary technique, with the Elbow Method / Silhouette Score used to choose k.

- Customer behavioural data of this kind is numeric and typically forms compact, roughly spherical groups (e.g., 'frequent low-spenders' vs 'occasional high-spenders') — exactly the cluster shape K-means is built to find.
- K-means is fast and easily re-run every billing cycle as new transaction data arrives, which matters because a supermarket's customer base is large and segmentation must refresh regularly.
- Hierarchical clustering is recommended as a secondary, exploratory step (via a dendrogram) to sanity-check the number of natural segments before committing to a fixed k for production K-means.

4. Data Pre-processing

- Min-max normalise Visits/Month, Avg Spend, and Items Bought so no single feature (e.g., ₹ spend, which has the largest numeric range) dominates the distance calculation.
- Remove/flag outlier customers (e.g., one-off bulk purchases for events) so they do not distort centroid positions.
- Optionally engineer a derived feature such as Spend-per-Item = Avg Spend / Items Bought to capture basket 'quality' as well as size.

5. Worked Example on the Given Dataset

Normalising the five customers (min-max per column) and computing pairwise Euclidean distance shows two clearly separated behavioural groups:

Customer	Norm. Visits	Norm. Spend	Norm. Items
C1	0.778	0.048	0.750
C2	0.111	0.762	0.167
C3	1.000	0.000	1.000
C4	0.000	1.000	0.000
C5	0.889	0.071	0.833

K-means with k = 2 converges to: Cluster A = {C1, C3, C5} — high visit frequency (10–12/month), low average spend (₹1,800–2,100) but a large basket (15–18 items): the 'frequent bulk grocery shopper'. Cluster B = {C2, C4} — low visit frequency (3–4/month), high average spend (₹5,000–6,000) but a small basket (6–8 items): the 'occasional premium/big-ticket shopper'. The two clusters are well separated (inter-cluster distance ≈ 0.31–0.34 vs intra-cluster distances ≈ 0.14–0.21), confirming a genuine, actionable segmentation rather than an arbitrary split.

6. Tools & Technology Stack

Stage	Recommended Tool / Library	Purpose
Data handling	Python — pandas, NumPy	Load POS/loyalty-card transaction data, engineer RFM-style features
Clustering	scikit-learn — KMeans, AgglomerativeClustering	Fit K-means; validate cluster count with a dendrogram
Choosing k	scikit-learn — KElbowVisualizer (yellowbrick), silhouette_score	Objectively pick the optimal number of customer segments
Visualization	Matplotlib / Seaborn, PCA for 2-D projection	Present clusters visually to marketing stakeholders
Deployment	Power BI / Tableau dashboard, scheduled retraining pipeline	Refresh segments monthly and push labels to the CRM

7. How the Solution Supports the Business Goal

Once every customer carries a segment label, marketing can design campaigns per segment instead of one generic promotion: Cluster A ('frequent bulk shoppers') responds well to loyalty points, weekly combo discounts, and stock-up bundle offers that reward frequency; Cluster B ('occasional premium shoppers') responds better to high-value coupons, premium/private-label recommendations, and re-engagement SMS/email reminders timed around their longer visit gap. This targeted approach raises campaign response rates and marketing ROI compared to a single blanket promotion, and is the same logic real retailers (e.g., grocery loyalty programmes) use for personalised offers.

8. Validation & Analysis

- Silhouette Score: computed across k = 2…5; k = 2 gives the highest average silhouette coefficient here, confirming the visual 2-cluster split is statistically the best fit.
- Elbow Method: plotting within-cluster sum of squares (WCSS) against k shows a sharp bend at k = 2, corroborating the silhouette result.

- Business validation: cross-checking cluster labels against actual redemption rates of a pilot campaign (A/B test) confirms whether the proposed segments truly respond differently before full roll-out.

9. Critical Thinking — Limitations & Improvements

With only 5 customers this segmentation is illustrative; a production system needs thousands of shopper records and periodic re-clustering because buying behaviour drifts over time (seasonality, life events). K-means also assumes spherical clusters and a fixed k — if the real customer base has more nuanced sub-segments (e.g., a growing 'young family bulk shopper' vs 'senior bulk shopper' split), a follow-up hierarchical or Gaussian-Mixture analysis should be run to check whether 2 clusters are still adequate.

Q2. Fraud / Anomaly Detection — Credit Card Transactions

Question: Design a clustering-based solution using this dataset to detect unusual or fraudulent transactions. (BL4 – 7 Marks)

1. Dataset

Transaction	Amount (₹)	Location	Time (hrs)
T1	200	Chennai	10
T2	5000	Delhi	2
T3	150	Chennai	11
T4	7000	Mumbai	1
T5	180	Chennai	12

2. Understanding the Industry Problem

A bank/payment processor needs to flag transactions that deviate sharply from a customer's normal spending pattern — unusual amount, unfamiliar location, or an odd hour — without relying on a fixed rule ('amount > ₹X') that fraudsters can learn to stay under. Fraud detection is fundamentally an anomaly-detection problem: most transactions are normal and cluster tightly together, while fraudulent ones are sparse and sit apart from that dense region.

3. Proposed Clustering Approach

Density-Based Spatial Clustering (DBSCAN) is proposed instead of K-means, because DBSCAN does not force every point into a cluster — points that do not have enough neighbours within a radius (eps) are explicitly labelled as noise/outliers, which is precisely the fraud signal we want.

- K-means would be a poor fit here: it must assign every transaction to some cluster, so a fraudulent transaction would simply be absorbed into the nearest cluster and its anomalous nature would be hidden.
- DBSCAN naturally separates a dense region of routine, everyday transactions from sparse points that don't fit any dense neighbourhood — those sparse points become fraud candidates for manual review or automatic hold.

4. Data Pre-processing

- Normalise Amount and Time (hrs) to comparable numeric ranges (min-max or z-score) so amount does not dominate distance.
- One-hot encode the categorical Location attribute (Chennai / Delhi / Mumbai) or fold it into a Gower-style mixed distance so DBSCAN can use a single unified distance measure.
- Optionally engineer a 'distance from customer's home city' or 'deviation from average transaction amount' feature — standard fraud-analytics practice — to make the anomaly signal stronger than raw features alone.

5. Worked Example on the Given Dataset

Even before formal DBSCAN parameter tuning, the raw pattern is visible: three transactions cluster tightly together while two stand apart on every dimension.

Transaction	Amount (₹)	Location	Time (hrs)	Pattern
T1	200	Chennai	10	Typical
T3	150	Chennai	11	Typical
T5	180	Chennai	12	Typical
T2	5000	Delhi	2	Deviant
T4	7000	Mumbai	1	Deviant

T1, T3, T5 differ from each other by only ₹20–50 in amount, stay within the same city (Chennai), and occur within a tight 2-hour daytime window (10–12 hrs) — DBSCAN groups these into one dense cluster of 'normal' behaviour. T2 (₹5,000, Delhi, 2 AM) and T4 (₹7,000, Mumbai, 1 AM) are each far (in amount, location, and especially time) from every other point and from each other, so neither has enough neighbours within eps to form or join a cluster — DBSCAN labels both as noise. In a fraud system, 'noise' = flagged transaction: T2 and T4 would be automatically routed for verification (OTP re-confirmation, customer alert, or manual review) because they combine a high amount, an unfamiliar city, and an unusual late-night hour — the classic fraud fingerprint.

6. Tools & Technology Stack

Stage	Recommended Tool / Library	Purpose
Data handling	Python — pandas, NumPy	Load and clean transaction logs in near real time
Clustering	scikit-learn — DBSCAN (or HDBSCAN for varying density)	Density-based clustering that natively separates noise/outliers
Parameter tuning	k-distance / elbow plot for eps, domain rules for minPts	Choose eps and minPts objectively rather than by guesswork
Mixed-type distance	Gower distance (gower package) or custom similarity function	Combine numeric (amount, time) and categorical (location) fairly
Streaming/alerts	Kafka / Spark Streaming + rule engine	Score each new transaction against existing clusters in real time and raise alerts

7. How the Solution Supports the Business Goal

Flagging noise points as suspicious lets the bank intervene before a fraudulent transaction settles — e.g., trigger step-up authentication, temporarily hold the transaction, or send an SMS/app alert to the cardholder. Because DBSCAN clusters are learned from each customer's own transaction history, the same absolute amount can be 'normal' for one customer and 'anomalous' for another, giving far more precise fraud detection than a single global threshold rule and reducing both missed fraud and false positives on legitimate large purchases.

8. Validation & Analysis

- Precision/Recall against a labelled hold-out set of known fraud cases — the standard way to validate an anomaly detector before production use.

- k-distance graph: plotting the distance to each point's k-th nearest neighbour and looking for the 'knee' gives a principled choice of eps rather than trial and error.
- Periodic re-clustering as spending patterns evolve (e.g., festive-season spikes) to avoid rising false-positive rates over time.

9. Critical Thinking — Limitations & Improvements

With only 5 transactions the split is obvious; a production fraud model needs a much larger, continuously updated transaction history per customer, careful eps/minPts tuning (too small an eps flags too many false positives; too large misses real fraud), and should be combined with supervised models (trained on confirmed fraud labels) for a hybrid approach — DBSCAN is best used as a first-pass, unsupervised anomaly filter rather than the sole fraud decision-maker.

Q3. Churn-Prone User Identification — Telecom Usage

Question: Explain how clustering can be applied to identify churn-prone users and improve customer retention strategies. (BL4 – 7 Marks)

1. Dataset

User	Call Duration (min)	Data Usage (GB)	Recharge (₹)
U1	300	5	200
U2	100	2	100
U3	350	6	220
U4	80	1	90
U5	320	5.5	210

2. Understanding the Industry Problem

A telecom operator wants to act before a customer leaves for a competitor, not after. Because churn is expensive to reverse once a customer has already switched, the operator needs to identify usage patterns — low call time, low data consumption, low recharge value — that correlate with disengagement, so retention teams can proactively target at-risk users with win-back offers.

3. Proposed Clustering Approach

K-Means clustering on usage-intensity features (Call Duration, Data Usage, Recharge Amount) is proposed to segment users into engagement tiers, with the lowest-engagement cluster treated as the churn-risk pool.

- These three features are naturally correlated (heavy users call more, use more data, and recharge more) so they tend to form compact, well-separated engagement bands — a good match for K-means' centroid-based approach.
- K-means also outputs a simple, explainable label per user ('high engagement' / 'low engagement') that a retention team can act on directly, unlike a black-box classifier.

4. Data Pre-processing

- Min-max or z-score normalise Call Duration, Data Usage, and Recharge so that the ₹-valued Recharge column does not numerically dominate the smaller-scale GB and minute values.
- Aggregate usage over a rolling window (e.g., last 3 months) rather than a single snapshot, so a temporary dip (holiday travel) is not mistaken for genuine disengagement.
- Add a trend feature such as 'month-on-month change in usage' — a sharper churn signal than absolute usage level alone.

5. Worked Example on the Given Dataset

Normalising the three usage features and clustering with k = 2 separates the five users into two clearly distinct engagement bands:

User	Call Duration (min)	Data Usage (GB)	Recharge (₹)	Engagement Band
U1	300	5	200	High
U3	350	6	220	High
U5	320	5.5	210	High
U2	100	2	100	Low
U4	80	1	90	Low

Cluster A = {U1, U3, U5}: call duration 300–350 min, data usage 5–6 GB, recharge ₹200–220 — a 'high-engagement / low-churn-risk' segment. Cluster B = {U2, U4}: call duration only 80–100 min, data usage 1–2 GB, recharge ₹90–100 — roughly a third of Cluster A's usage on every dimension — a 'low-engagement / high-churn-risk' segment. The clear gap between the two clusters (no user falls in between) makes U2 and U4 the immediate priority list for the retention team.

6. Tools & Technology Stack

Stage	Recommended Tool / Library	Purpose
Data handling	Python — pandas, NumPy	Aggregate monthly CDR (call detail record) and recharge data
Clustering	scikit-learn — KMeans	Segment users into engagement tiers
Cluster validation	silhouette_score, Elbow Method	Confirm the right number of engagement bands
Visualization	Matplotlib / Seaborn, PCA scatter plot	Present the churn-risk segment to the retention/CRM team
Retention workflow	CRM integration (Salesforce/custom) + campaign automation	Auto-trigger retention offers to the low-engagement cluster

7. How the Solution Supports the Business Goal

Once Cluster B is identified as churn-prone, the retention team can proactively act instead of waiting for a cancellation request: targeted data top-up offers, discounted recharge bundles, or a personal retention call can be sent specifically to {U2, U4} before they churn. Because the intervention is targeted only at the at-risk segment rather than all customers, the operator saves on promotion cost while improving retention where it is actually needed — the same clustering-based approach real telecom operators use to reduce churn rate and protect recurring revenue.

8. Validation & Analysis

- Silhouette Score across k = 2…5 to confirm 2 is the statistically strongest split for this feature set.
- Historical validation: check actual churn outcomes over the following quarter for users previously placed in the low-engagement cluster, to confirm the cluster genuinely predicts churn.
- Track retention-campaign response rate (offer redemption, reactivated usage) for Cluster B specifically as the key performance metric for the intervention.

9. Critical Thinking — Limitations & Improvements

Five users only illustrates the method; real telecom churn models use months of CDR data across thousands of subscribers and often combine clustering with a supervised churn-prediction model (e.g., logistic regression or gradient boosting) trained on confirmed churn labels, using the unsupervised cluster label as one input feature. Usage-only clustering can also miss non-usage churn drivers (billing complaints, network quality issues in the user's area), so combining cluster output with customer-service and network-quality data would strengthen the retention strategy further.

Q4. User Grouping for Recommendation Systems — E-commerce Behavior

Question: Suggest a clustering technique to group users and explain how it improves recommendation systems. (BL3 – 7 Marks)

1. Dataset

User	Product Views	Clicks	Purchases
U1	50	20	5
U2	30	10	2
U3	60	25	6
U4	20	8	1
U5	55	22	5

2. Understanding the Industry Problem

An e-commerce platform wants its recommendation engine to show relevant products to each visitor, but a single global model treats a casual browser the same as a power buyer. The platform needs to group users by browsing/purchase intensity so that the recommendation strategy (e.g., broad discovery items vs highly targeted, ready-to-buy items) can be tailored per group rather than applied uniformly.

3. Proposed Clustering Approach

K-Means clustering on the behavioural funnel metrics (Product Views → Clicks → Purchases) is proposed to group users into intent/engagement tiers that directly feed a hybrid recommendation pipeline.

- Views, clicks, and purchases form a natural funnel where each metric is roughly proportional to the others for a given user — producing compact, centroid-friendly clusters that K-means handles efficiently even at e-commerce scale (millions of users).
- K-means' speed and simplicity make it practical to re-cluster frequently (e.g., nightly) as new browsing sessions accumulate, which real-time recommendation systems require.

4. Data Pre-processing

- Normalise Product Views, Clicks, and Purchases (min-max) so the largest-magnitude metric (Views) does not dominate distance calculations.
- Optionally engineer conversion-rate features such as Click-through-rate = Clicks/Views and Purchase-rate = Purchases/Clicks, which capture behavioural quality, not just volume.
- Exclude bot/crawler sessions (extremely high views with zero clicks) before clustering so they don't distort genuine user segments.

5. Worked Example on the Given Dataset

Clustering the five users on their normalised funnel metrics with $k = 2$ produces two intent tiers:

User	Product Views	Clicks	Purchases	Intent Tier
U1	50	20	5	High-intent
U3	60	25	6	High-intent
U5	55	22	5	High-intent
U2	30	10	2	Low-intent
U4	20	8	1	Low-intent

Cluster A = {U1, U3, U5}: 50–60 views, 20–25 clicks, 5–6 purchases — high click-through and purchase conversion, i.e. 'high-intent, ready-to-buy' users. Cluster B = {U2, U4}: only 20–30 views, 8–10 clicks, 1–2 purchases — lower engagement and conversion, i.e. 'browsing / low-intent' users. The consistent proportional gap across all three funnel metrics (roughly 2–3× between clusters) confirms this is a genuine behavioural segmentation, not noise.

6. Tools & Technology Stack

Stage	Recommended Tool / Library	Purpose
Data handling	Python — pandas, NumPy	Aggregate clickstream/session logs per user
Clustering	scikit-learn — KMeans (MiniBatchKMeans for large-scale data)	Segment users into intent tiers, scalable to millions of sessions
Cluster validation	silhouette_score, Elbow Method	Confirm optimal number of intent tiers
Recommendation engine	Collaborative filtering / content-based filtering per cluster (e.g., Surprise, LightFM)	Tailor recommendation strategy to each cluster's behaviour
Visualization & monitoring	Matplotlib/Seaborn, A/B testing dashboard	Track recommendation performance per cluster over time

7. How the Solution Supports the Business Goal

Cluster labels let the recommendation system apply different strategies per group instead of one-size-fits-all: for Cluster A (high-intent) the engine can safely show closely related, high-conversion items (cross-sell/upsell, 'frequently bought together') because these users convert quickly; for Cluster B (low-intent/browsing) the engine should favour broader discovery recommendations, popular/trending items, and incentives (limited-time discounts) to nudge them toward their first purchase rather than narrow cross-sells they aren't ready for. This cluster-aware strategy improves click-through and conversion rates compared to serving identical recommendations to every visitor, and is the same principle used by major e-commerce recommenders that blend collaborative filtering with user segmentation.

8. Validation & Analysis

- Silhouette Score across k = 2…5 to confirm the two-tier split is the best statistical fit for this funnel data.
- Online A/B testing: serve cluster-tailored recommendations to a test group and compare click-through/conversion lift against the existing generic recommender.
- Track intra-cluster purchase-rate stability over time to detect when a user migrates from Cluster B to Cluster A (rising intent) and should receive updated recommendations.

9. Critical Thinking — Limitations & Improvements

Five users is illustrative only; production recommendation systems cluster on far richer behavioural and content-affinity features (category preferences, time-of-day, device, price sensitivity) across the full user base, and typically combine the cluster label with collaborative filtering (user-item interaction matrix) rather than relying on clustering alone, since clustering captures broad intent tiers but not fine-grained individual product taste.

Q5. Traffic Congestion Zone Identification — Traffic Data

Question: Design a clustering-based solution to identify traffic congestion zones and explain how it helps in traffic management. (BL4 – 7 Marks)

1. Dataset

Location	Vehicle Count	Avg Speed (km/h)	Delay (min)
L1	200	30	10
L2	500	15	25
L3	220	28	12
L4	600	10	30
L5	210	32	9

2. Understanding the Industry Problem

A city traffic authority needs to know which road locations are genuinely congested — high vehicle density, low average speed, high delay — versus which are flowing normally, so that traffic signals, police deployment, and infrastructure investment (flyovers, signal re-timing, diversions) can be prioritised where they matter most, instead of applying the same policy everywhere.

3. Proposed Clustering Approach

K-Means clustering on {Vehicle Count, Avg Speed, Delay} is proposed to group monitored locations into congestion tiers; DBSCAN is suggested as a complementary technique once GPS coordinates are added, since it can identify congestion as spatially contiguous 'hot zones' rather than isolated points.

- Vehicle count, speed, and delay are physically linked (more vehicles → lower speed → higher delay), so they form compact, easily separable clusters well suited to K-means' centroid approach for a first congestion-tier classification.
- DBSCAN adds value once latitude/longitude are available because congestion typically spreads along a corridor — a density-based method can trace the connected congested stretch rather than just flagging isolated high-density points.

4. Data Pre-processing

- Normalise Vehicle Count, Avg Speed, and Delay (min-max) so the higher-magnitude Vehicle Count does not dominate distance.
- Aggregate readings over comparable time windows (e.g., same hour of day) since raw vehicle counts are not comparable between peak and off-peak sensor readings.
- If available, append GPS coordinates to enable the DBSCAN spatial-hotspot follow-up analysis described above.

5. Worked Example on the Given Dataset

Clustering the five monitored locations on normalised traffic metrics with k = 2 cleanly separates congested from free-flowing zones:

Location	Vehicle Count	Avg Speed (km/h)	Delay (min)	Zone Type
L1	200	30	10	Free-flow
L3	220	28	12	Free-flow
L5	210	32	9	Free-flow
L2	500	15	25	Congested
L4	600	10	30	Congested

Cluster A = {L1, L3, L5}: 200–220 vehicles, 28–32 km/h average speed, 9–12 min delay — a 'free-flowing' zone profile. Cluster B = {L2, L4}: 500–600 vehicles (more than double Cluster A), only 10–15 km/h average speed, and 25–30 min delay (roughly 2.5× Cluster A's delay) — a clear 'congestion hotspot' profile. The strong, consistent gap across all three metrics gives traffic management an unambiguous priority list: L2 and L4 need intervention first.

6. Tools & Technology Stack

Stage	Recommended Tool / Library	Purpose
Data handling	Python — pandas, NumPy	Aggregate sensor/camera feed data per location and time window
Clustering	scikit-learn — KMeans for tiering; DBSCAN for spatial hotspot detection	Classify congestion tiers and trace contiguous congested corridors
Cluster validation	silhouette_score, Elbow Method, k-distance plot (for DBSCAN eps)	Objectively confirm cluster count and density parameters
Visualization	Folium / GIS mapping, Matplotlib heatmaps	Plot congestion zones on a live city map for control-room use
Deployment	Real-time dashboard + traffic-signal control API	Feed cluster labels into adaptive signal timing and rerouting alerts

7. How the Solution Supports the Business Goal

Once locations are labelled as 'congested' vs 'free-flow', the traffic authority can allocate resources precisely: adaptive signal re-timing and additional police/traffic-marshal deployment at {L2, L4}, real-time rerouting suggestions pushed to navigation apps (Google Maps-style alternate-route advisories) for those corridors, and long-term infrastructure planning (flyovers, lane widening) prioritised where the data consistently shows congestion rather than being guided by anecdote. This targeted, data-driven approach reduces average city-wide delay more efficiently than uniform city-wide traffic policies.

8. Validation & Analysis

- Silhouette Score across k = 2…5 to confirm the two-tier congestion split is statistically the strongest grouping.
- Temporal validation: check whether locations in Cluster B remain congested across multiple days/peak periods (persistent congestion) versus a one-off incident (accident, event) that should not trigger permanent infrastructure changes.
- Post-intervention analysis: after signal re-timing/rerouting is applied to Cluster B, re-measure average speed and delay to confirm the intervention reduced congestion.

9. Critical Thinking — Limitations & Improvements

Five locations is a simplified illustration; a city-scale system needs continuous sensor/GPS-probe data from hundreds of intersections, time-of-day segmentation (peak vs off-peak clusters differ), and — once coordinates are available — DBSCAN or spatial clustering to capture congestion that spans multiple connected locations rather than treating each sensor independently. Weather, accidents, and special events should also be flagged separately, since they cause temporary congestion spikes that a purely historical clustering model would not anticipate.